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ITAI 1371
Midterm Outline

Interest Area:

Using Machine Learning to Detect Malicious Network Traffic

Start With You

Chosen Area: Cybersecurity / Technology

I'm interested in how machine learning can help detect cyber threats faster and more accurately than manual analysis.

Turn Your Interest Into ML Project Ideas

1. **Prediction:**
Can ML predict whether network traffic is malicious using packet-level features?
2. **Classification:**
Can ML classify network traffic into benign vs malicious categories?
3. **Detection:**
Can I detect cyber attacks from network traffic data?

Chosen idea: Classification (benign vs malicious)

State of the Art (What Others Are Doing)

Research Paper

- **Paper:** "Machine Learning-Based Intrusion Detection Systems"
- **What are they solving?**
Detecting network intrusions automatically using ML.
- **What data did they use?**
Public intrusion datasets like NSL-KDD.
- **What ML method did they use?**
Decision Trees, Random Forests, and SVM.

- **Important result:**
Random Forest achieved high detection accuracy compared to traditional methods.

Trusted Article / Blog

- **Source:** IBM Security Blog on AI for Threat Detection
 - **What are they solving?**
Reducing time to detect cyber attacks.
 - **What data did they use?**
Network logs and traffic telemetry.
 - **What ML method did they use?**
Supervised classification and anomaly detection.
 - **Important result:**
AI reduces breach detection time significantly.
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Big Companies Using This

Companies Chosen

1. Microsoft
2. IBM
3. Google

Company	What are they building?	How ML helps	Interesting detail
Microsoft	Defender for Endpoint	Detects malicious behavior	Uses ML for real-time detection
IBM	QRadar & Watson	Threat correlation	Reduces breach costs
Google	Chronicle Security	Large-scale log analysis	Handles petabytes of data

Shrink Topic to Course Level

1. **Dataset:**
NSL-KDD or UNSW-NB15 (public intrusion datasets)
2. **Input (Features):**
Packet size, protocol type, duration, service, flags
3. **Output (Target):**
Malicious (1) or Benign (0)
4. **Algorithm Type:**
Classification
5. **Models to Try:**

- Logistic Regression
 - Decision Tree
 - Random Forest
 - K-Nearest Neighbors
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Abstract

This project explores the use of machine learning for detecting malicious network traffic in cybersecurity environments. As cyber-attacks continue to grow in volume and complexity, traditional rule-based detection systems struggle to scale effectively. Recent research and industry solutions from companies such as Microsoft, IBM, and Google demonstrate that ML-based intrusion detection can significantly improve detection accuracy and response times. Using a publicly available network intrusion dataset, this project will build and evaluate several supervised classification models to distinguish between benign and malicious traffic. The results will be compared against existing research findings to assess performance and highlight the practicality of ML-based threat detection at an introductory level.